Project Gaja-Saviya

ECOSYSTEM DESIGNED TO MITIGATE THE HUMAN-ELEPHANT CONFLICT

R26-IT-136

RUMBLE

TEAM CONTRIBUTIONS



Team Member :
W.M.R.A.B.Wijesooriya
Component : Behavioral Analysis System
IT22249098



Team Member :
Dhananjaya A G G
Component : Fence Monitoring System
IT22145734



Team Member
:GUNATHILAKE D R G L
Component : Path-Guard
IT22544636



Team Leader :
Saisuhani L
Component : Conflict Prediction System
IT22091598

Comments from Panel and the Actions we took

Component	Proposal Comment	Action Taken / Technical Resolution
Project Wide	Initial 'Wellness App' lacked feasible data access and had an undefined research scope."	**Strategic Topic Pivot:** Developed **Gaja-Saviya**. Shifted focus to Human-Elephant Conflict where we secured accessible data and defined a high-impact IoT-ML scope.
C1: Fence Monitoring System	"How do you know if the fence is actually broken or just has a technical fault?"	Integrated **Differential Threshold Logic** to distinguish between a wire cut (Breach) and a branch contact (Fault) based on voltage/current drop patterns.
C2: Acoustic-Intent	"How can you ensure your ML model stays accurate if the environmental noise changes, or if local elephants have different vocal frequencies?"	As app collects more "Angry Elephant" sounds on the SD card, ML model gets smarter. implement **OTA Updates** to update the ESP32's threshold logic or firmware wirelessly.
C3: Danger Prediction	"Prediction cannot rely on historical data alone; it needs to be context-aware."	Built an **Ensemble Learning Model** that merges historical DWC records with live meteorological data (rainfall) and agricultural crop harvest cycles.
C4: Path-Guard	"Reporting is restricted to on-sight persons; users not near the sighting cannot contribute, limiting the safety data pool."	**Implemented Global Crowd-Sourcing:** Added a "Remote Reporting" feature allowing any user to pin a sighting. Integrated a **ML-based Trust Engine** to verify remote reports based on proximity and user reputation scores.

Project Gaja-Saviya
AI-Powered Human–Elephant Conflict Mitigation System
Overall System Diagram

1. IOT LAYER – DATA ACQUISITION

Electric Fence Signal Sensors

- Voltage / Current Pulses
- Electrical Disturbances

Acoustic Sensors

- Environmental Sounds
- Elephant Vocalizations

IoT Sentinel Node (ESP32)

- Data Collection
- Time Synchronization
- Secure Transmission

2. AI PROCESSING LAYER (MODALITY–SPECIFIC ANALYSIS)

Component 1 – Intelligent Fence Signal Interpreter (LSTM)

Input: Raw Electrical Signals

Preprocessing
- Denoising
- Segmentation
- Normalization

LSTM Model

Output
- Disturbance Type
- Confidence Score
- Timestamp

Component 2 – Bio-Acoustic Behaviour Analyst (CNN)

Input: Audio Signals

Preprocessing
- Framing
- Mel-Spectrogram
- Normalization

CNN Model

Output
- Behaviour Label
- Confidence Score
- Timestamp

3. CROSS-MODAL SENSOR FUSION LAYER

Decision Making & Threat Validation

From Component 1
- Disturbance Type
- Confidence Score
- Timestamp

From Component 2
- Behaviour Label
- Confidence Score
- Timestamp

1. **Timestamp Synchronization** — Match events in same time window
2. **Fusion Engine (Rule / Bayesian Decision Fusion)** — Combine predictions from both modalities
3. **Joint Confidence Calculation** — Compute final confidence score
4. **Threat Validation (Real Threat or False Alert)** — Validate using thresholds & rules
5. **Duplicate Alert Filter** — Remove repeated alerts in short intervals

FUSION OUTPUT
- Validated Threat Status
- Severity Level
- Joint Confidence Score
- Location (Node ID)
- Timestamp

REAL THREAT CONFIRMED

4. DATA LAYER (Storage & Management)

Database (MySQL / MongoDB)
- Alerts
- Sensor Data
- User Reports
- System Logs

Data Access API

API

FAST API

5. APPLICATION LAYER – USER INTERFACE & DECISION SUPPORT

Mobile Application
- Real-time Alerts
- Elephant Locations
- Safe Routes
- Incident Reporting
- Notifications

Dynamic Safe Routing
- Risk Map
- Safe Route Generation
- Avoid High-Risk Zones

Community & Trust Platform
- User Reports
- Trust Score
- Verified Sightings

Users

Farmers / Local Communities

Wildlife Officers / Authorities

SYSTEM BENEFITS

- Accurate Elephant Intrusion Detection
- Reduced False Alarms
- Real-time Decision Support
- Safe Navigation Guidance
- Strong Community Awareness

LEGEND

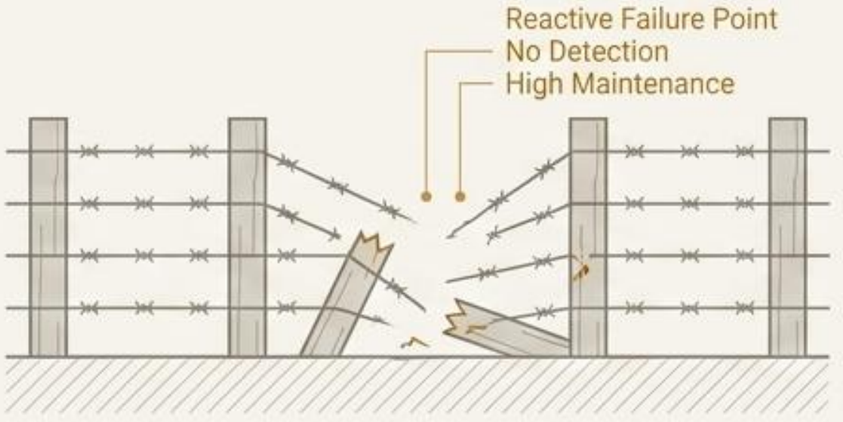
→ Data Flow

- - - → Internal Processing / Decision Flow

Gaja-Saviya

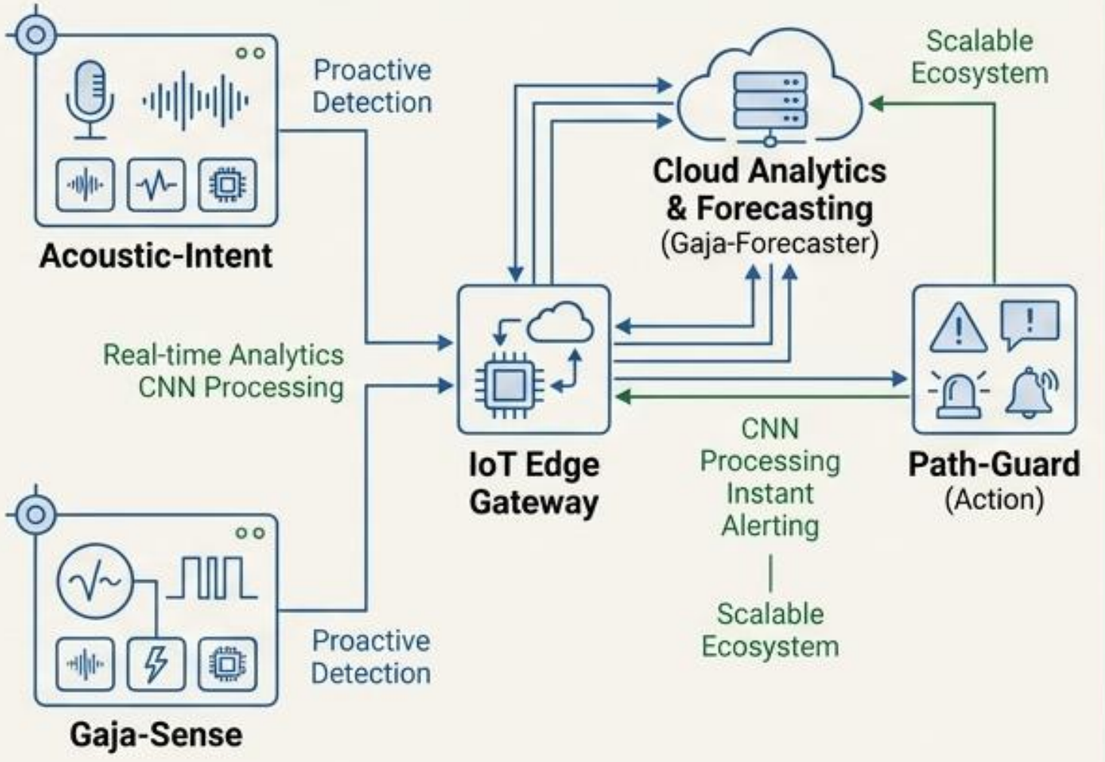
An Integrated AI-IoT Ecosystem for Proactive Human-Elephant Conflict Mitigation

Static Barrier Concept



Transformation

Dynamic Intelligence Network

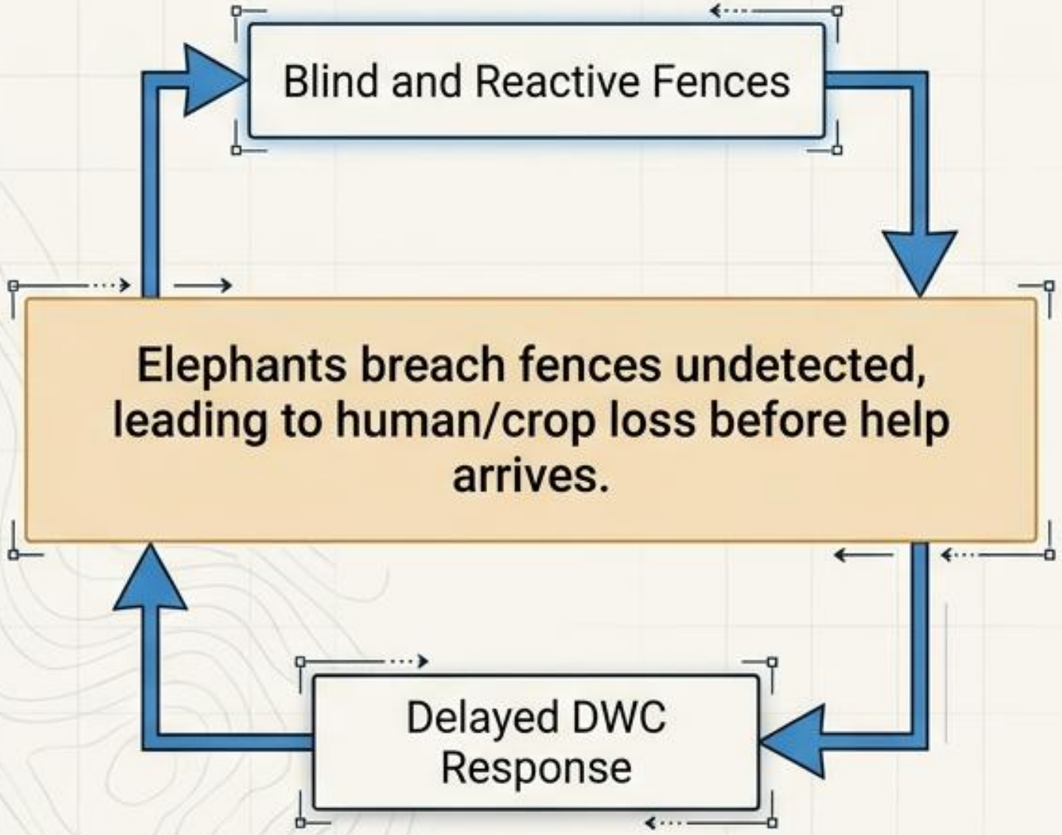


Proposal Feedback Pivot

Transitioned from a localized hardware solution to a dynamic network through deliberate integration of CNN architectures and IoT Edge Gateway logic.

The Fatal Flaw of Blind and Reactive Defenses

Cycle of Loss



Target Beneficiaries



Primary: Rural Farmers in HEC Zones

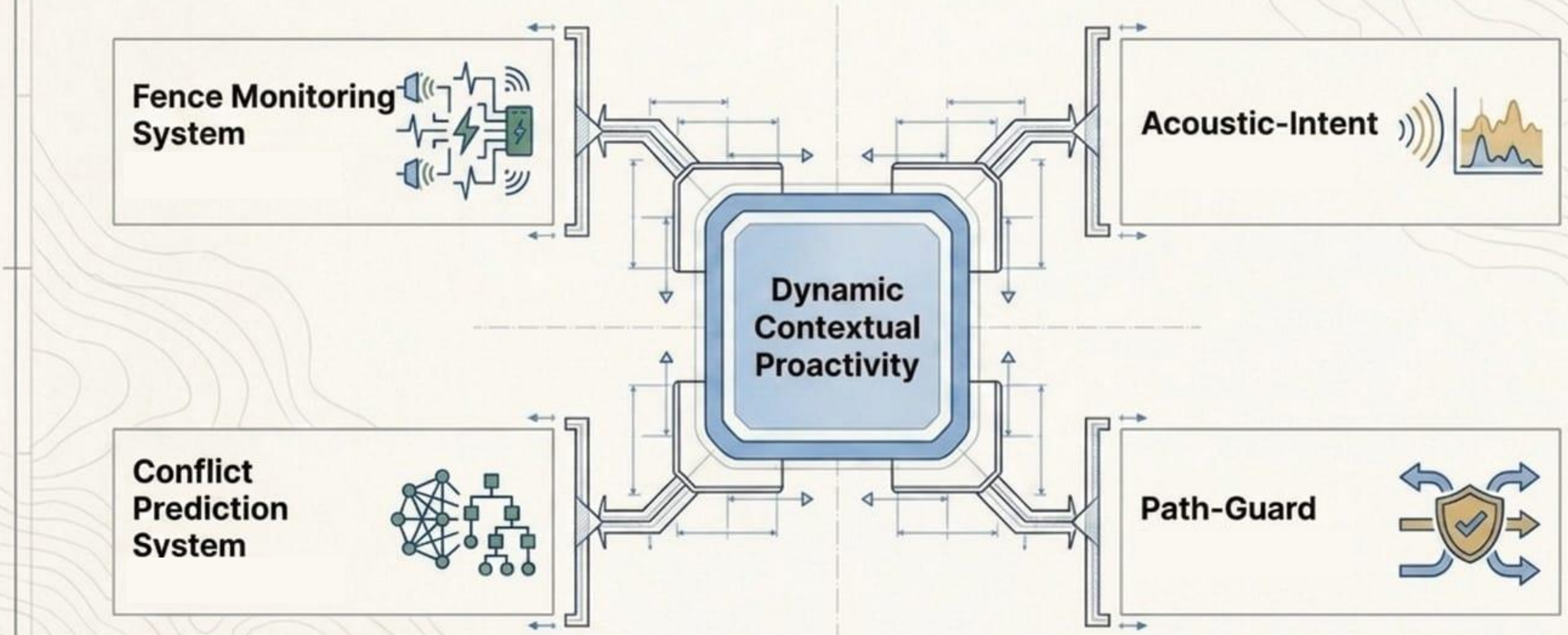
Focus: Crop protection and physical safety.



Secondary: Village Communities & DWC

Focus: Broad-scale alerts and resource allocation.

Ecosystem Architecture: Dynamic Contextual Proactivity



SNAPS FROM THE FIELD VISITS



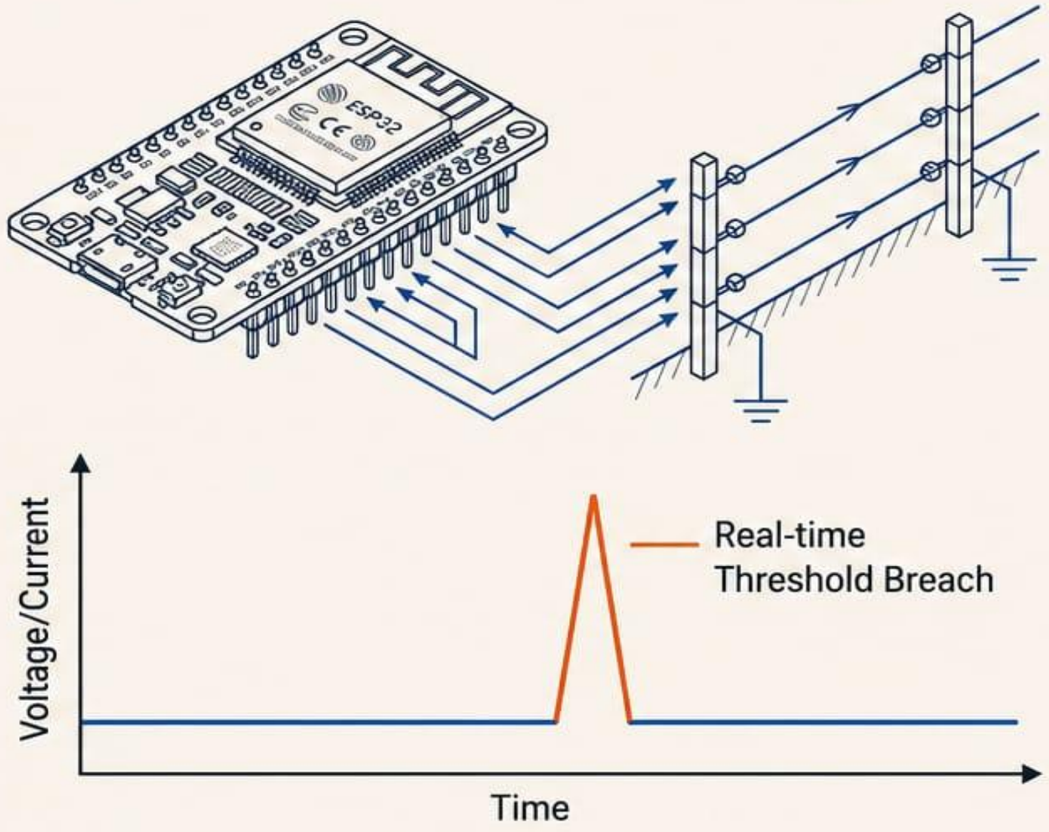




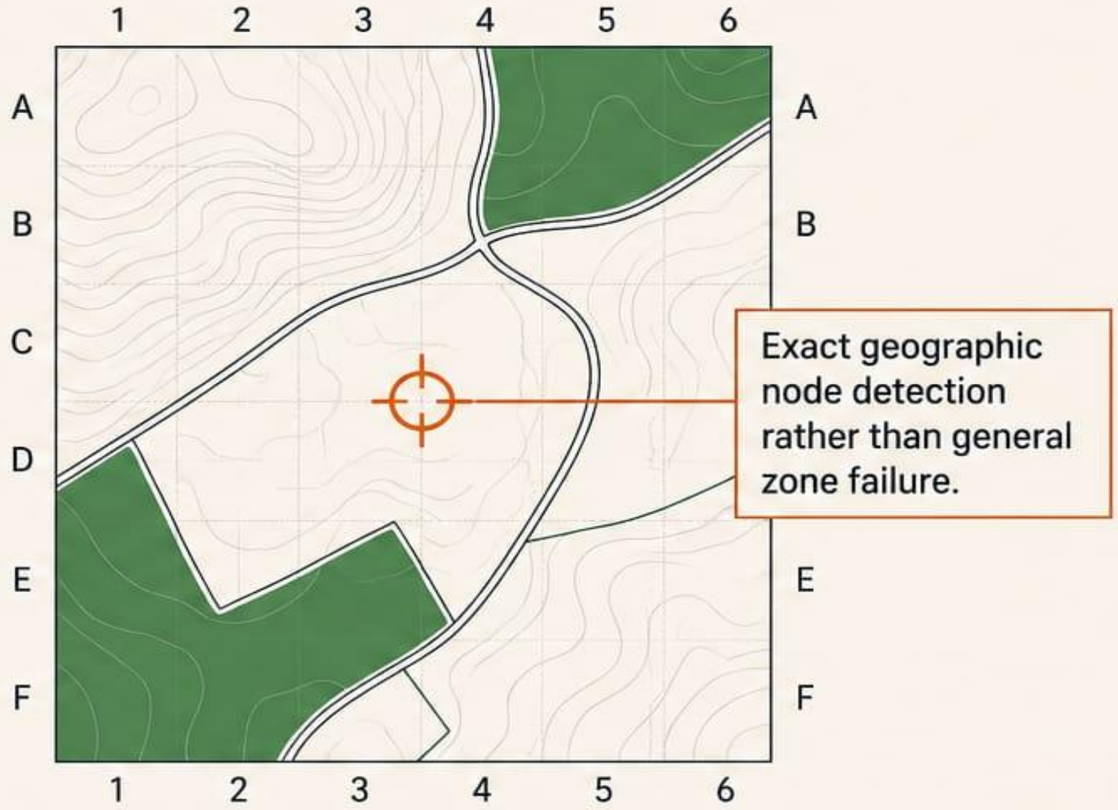
Real-Time Edge Monitoring



Hardware Methodology



Instant Fault Localization



IT22145734 | Dhananjaya A G G | R26-IT-136

Component 01: Fence Monitoring System



Sub-Problem:
Replacing easily compromised, reactive barriers with proactive, continuous sensing.

User Requirements:

1. **Low-latency alerts**
2. **Real-time breach detection**
3. **High reliability in harsh conditions**

Design Excellence:
Implementation of precise **Threshold Logic** to filter environmental noise from actual structural breaches.

User Feedback:
‘**94% Satisfaction** — Finally, we know immediately when the perimeter is tested, not just after it breaks.’

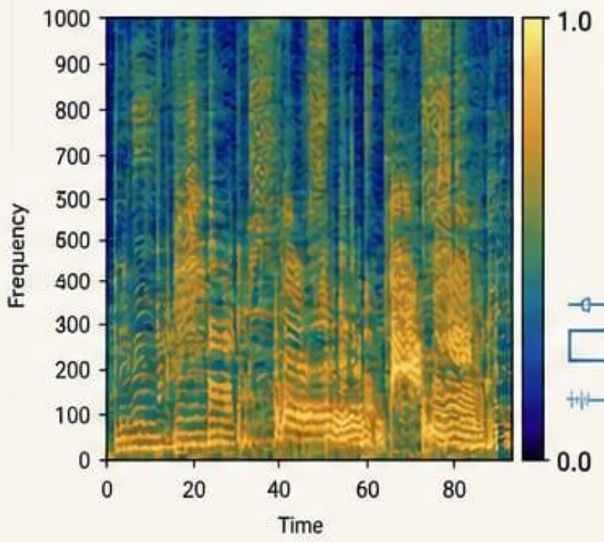
IT22145734 |Dhananjaya A G G| R26-IT-136

Acoustic-Intent: Dual-Stage Audio Classification



Mel-Spectrogram

CNN Processing



Raw Audio Transformation

Stage 1: Is it an elephant?

Yes

No

Stage 2: Behavioral State Recognition

Agonistic (Angry/Threatening)

Social (Calm)

IT22249098 |W.M.R.A.B.Wijesooriya| R26-IT-136

Component 02: Acoustic Intent Classification

Sub-Problem:
Moving beyond simple presence detection to understanding complex animal behavior and aggression levels.

User Requirements:
1. Behavioral confirmation
2. False-positive reduction
3. Early warning before visual contact

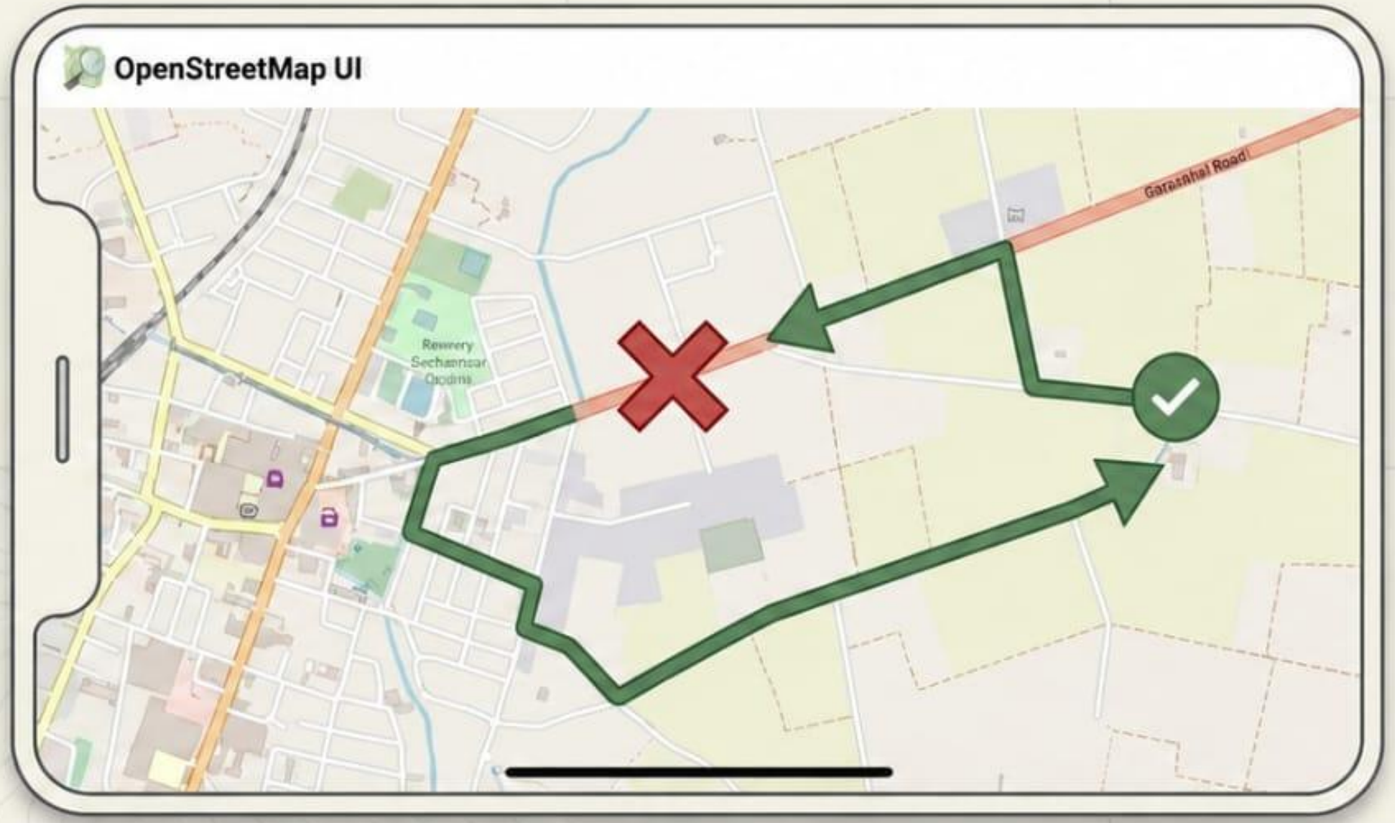


Design Excellence:
Developed a proprietary Two-Stage CNN to isolate elephant vocalizations and classify intent (e.g., foraging foraging vs. distress/aggression).

User Feedback:
“92% Satisfaction — Knowing how the elephant is acting completely changes how we respond.”

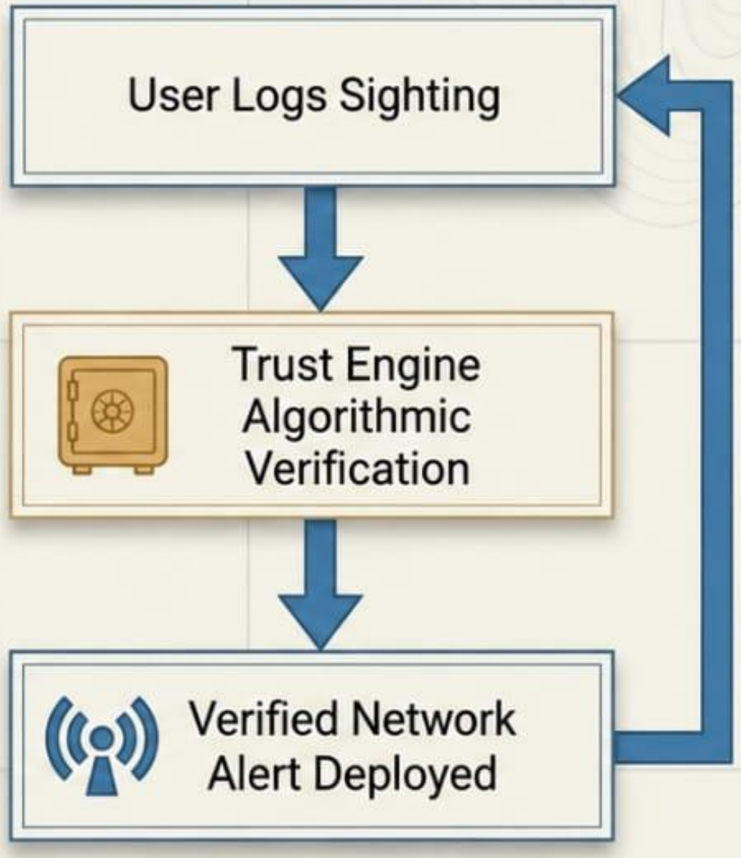
Path-Guard: Algorithmic Routing & Community Verification





Custom Weighted Dijkstra's Algorithm calculates dynamic safe detours.

Crowdsourced Reporting loop



Component 04: Path-Guard Dynamic Routing

Sub-Problem: Mitigating deadly surprise encounters during nighttime travel through dynamic, context-aware navigation.

User Requirements:

1. Instant rerouting
2. Offline-capable edge alerts
3. Safe-passage guarantees

Design Excellence:
Utilization of a Weighted Dijkstra Algorithm, dynamically adjusting path safety weights based on live sensor inputs.

User Feedback:
'96% Satisfaction — The navigation helps me find safe paths at night without fear.'



IT22544636 |GUNATHILAKE D R G L| R26-IT-136

Ensemble Learning for Danger Prediction



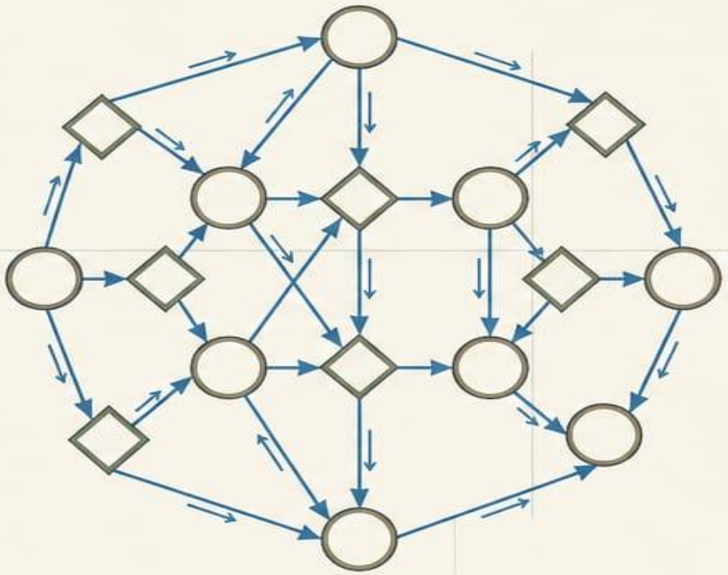
Historical DWC Data

Live Weather Patterns

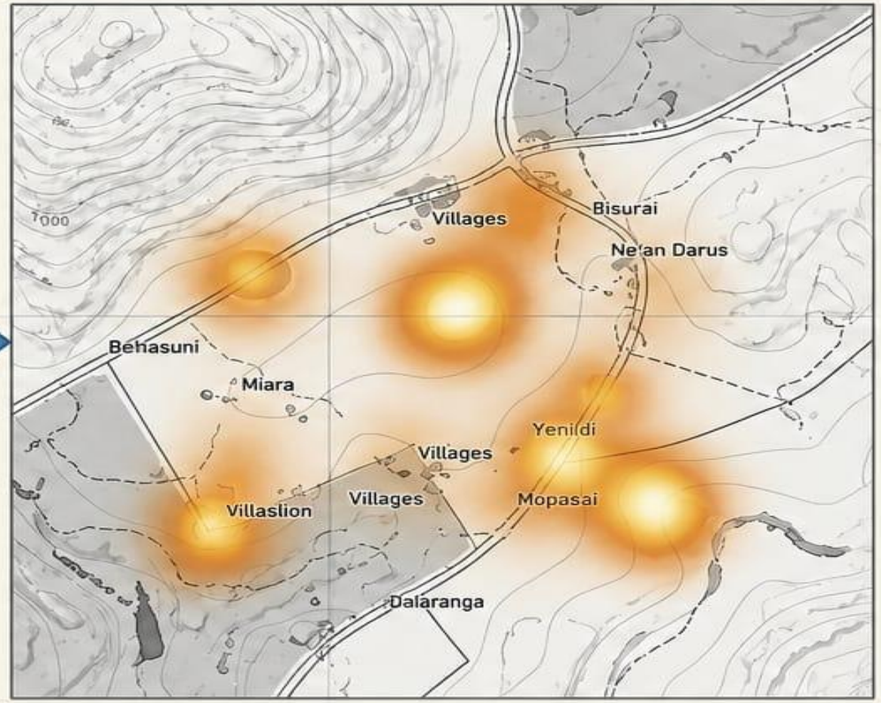
Crop Harvest Cycles

Ensemble Learning Engine

Random Forest Model



Predicts “High Danger” days



Component 03: Conflict Prediction System

Sub-Problem:
Shifting the paradigm from responding to immediate threats to forecasting potential conflict zones in advance.

User Requirements:
1. Predictive zoning
2. Resource allocation guidance
3. Integration of historical & real-time data



Design Excellence:
Advanced machine learning models synthesizing spatial-temporal data to generate high-probability conflict maps.

User Feedback:
‘91% Satisfaction — DWC officers can now pre-deploy resources before the elephants even approach the villages.”

Gaja-Saviya – Market Analysis & User Persona



Primary User: Rural Smallholder Farmer (e.g., Gunapala from Ampara)

Needs: Real-time alerts during crop guarding and safe navigation for night travel.



Secondary User: DWC Field Officer

Needs: Accurate data on fence breach locations and elephant behavioral trends.



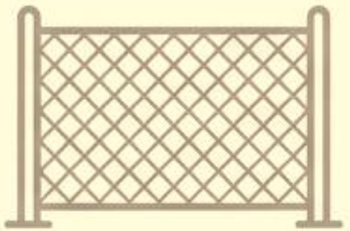
TAM (Total Addressable Market): 131 DS Divisions in Sri Lanka categorized as HEC-affected.

SAM (Serviceable Addressable Market): 5,000+ localized agricultural cooperatives.

Gaja-Saviya – The Unique Proactive Advantage

From Static Walls to Dynamic Intelligence.

Left Column: Traditional Defenses (Blind)



- Unable to assess threat.
- Unverified reports.
- No routing assistance.

Right Column: The Gaja-Saviya Intelligence Grid (Aware)



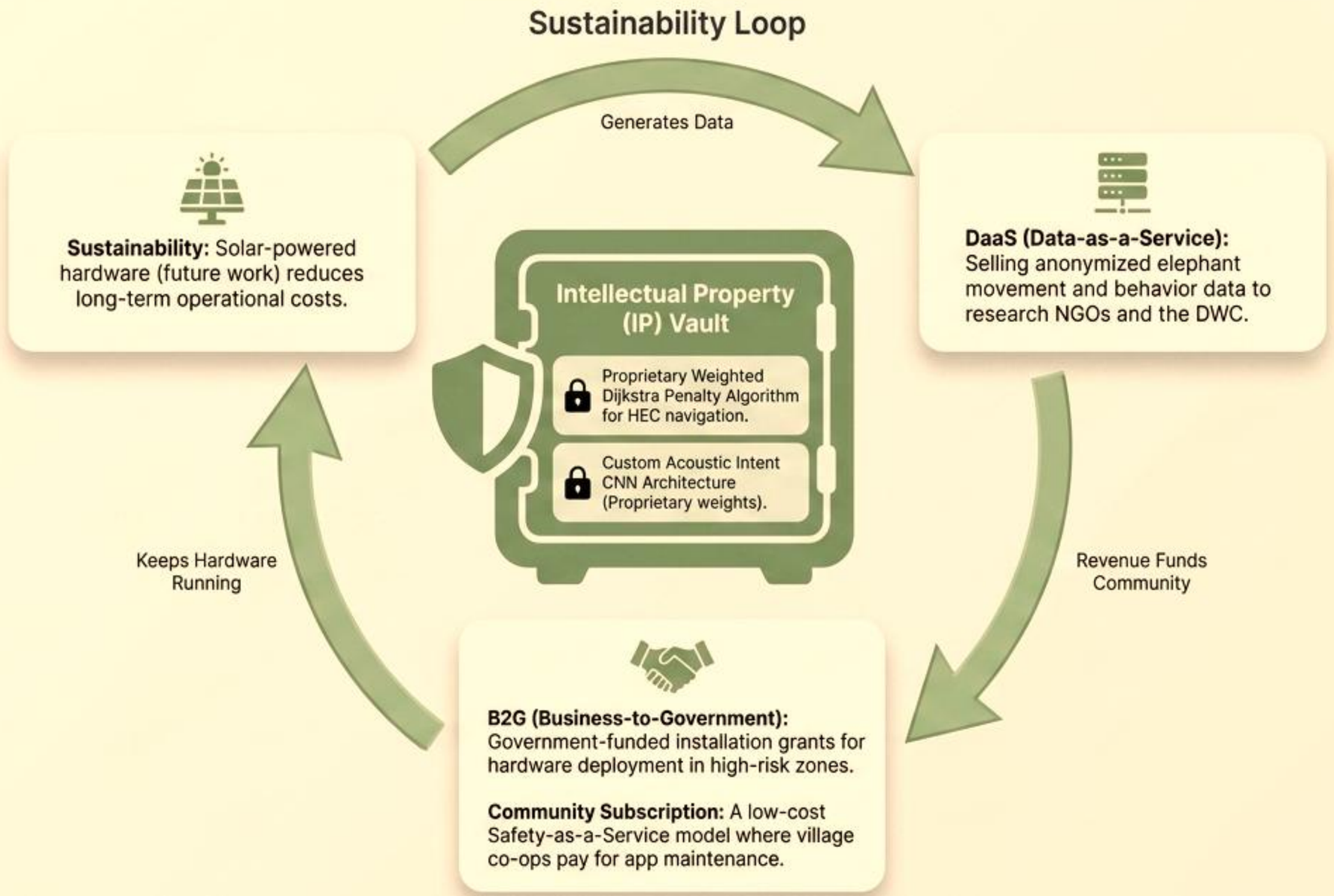
- **Contextual AI:** Differentiates between a calm herd and an aggressive lone bull (Component 2).
- **Trust-Engine Verification:** Crowdsourcing that filters out false reports based on proximity logic.
- **Dynamic Routing:** Real-time path recalculation using the Penalty Weight Dijkstra algorithm (Component 4).

Before vs. After Comparative Anatomy Matrix

Algorithmic Pathing



Gaja-Saviya – Financial Sustainability & IP





THANK YOU

RUMBLE